



Trusted by **25,000+** students in
Linux, Red Hat, Cloud, DevOps & AI

The Complete DevOps, MLOps & AI Engineering Program

The **Only** Program You'll Ever Need



280
HOURS



19
PHASES



15+
PROJECTS



DevOps Foundations + ML Operations + AI Engineering.
One Path. **Real Impact.**

About the Complete DevOps, MLOps & AI Engineering Program

GRRAS Solutions is India's premier training institute for Linux, Red Hat Certification, Cloud Infrastructure, DevOps, and now, Applied AI Engineering. The MLOps Engineering Program merges our flagship 180-hour DevOps curriculum with our 100-hour Machine Learning & Generative AI track into a single, sequential, 280-hour career path — built for engineers who want to own infrastructure *and* the AI systems running on top of it.

25,000+

Engineers Trained

Across Linux, Red Hat, Cloud & DevOps

19

Sequential Phases

DevOps foundations → Generative & Agentic AI

100%

Practical Labs

Real terminals, real clusters, real APIs

280 Hrs

2026-Ready

One program, two disciplines, one engineer

Why Merge DevOps & AI Into One Program?

✓

Because in 2026 hiring, they are no longer separate skill sets — job postings for “MLOps Engineer,” “AI Infrastructure Engineer,” and “Platform Engineer” all list the same toolchain: Linux, Docker, Kubernetes, Terraform, CI/CD — plus LLMs, RAG, and agents.

✓

Because every AI system eventually needs to be deployed, containerized, monitored, and kept running — that's DevOps work, just applied to models and agents instead of web apps.

✓

Because learning them together, in the right order, means every AI lab in this program reuses infrastructure you've already built — not toy examples on someone else's laptop.

✓

Because it produces one outcome the market is short on: an engineer who is equally credible running a Kubernetes cluster and shipping a production LLM application on it.

Why Learn MLOps at GRRAS?

✓

25,000+ students already trained in Linux, Red Hat, Cloud & DevOps — this program is the natural next step, not a cold start.

✓

Same trusted instructors and hands-on, lab-first teaching style known across top IT hubs.

✓

Curriculum designed specifically for infrastructure engineers — not a generic data-science bootcamp with DevOps bolted on.

✓

Every lab, from Linux hardening to agentic AI, is infrastructure-flavored: log analysis, incident response, runbook automation, and now, AI-augmented operations.

✓

One single, sequential 280-hour path — eliminating scattered YouTube tutorials, outdated courses, and disconnected bootcamps.

THE GRRAS GUARANTEE

Zero fluff. No infrastructure without production context, and no AI without the infrastructure to run it on. Every lab is built for real engineering work.

Index & Program Structure (1/2)

280-Hour, 19-Phase Sequential Journey — from Linux fundamentals through Generative & Agentic AI.

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P.17	Phase 11 — CI/CD with Jenkins or GitHub Actions Automated build, test, and deploy pipelines	✓
P.18	Phase 12 — Monitoring with Prometheus & Grafana Real-time observability, the bridge into AIOps	✓

Index & Program Structure (2/2)

Continued — Generative & Agentic AI, capstones, and career outcomes.

SECTION III — GENERATIVE & AGENTIC AI (100 HRS)

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Phase 13 — Data Science & ML Foundations
Python for AI, EDA, and classic ML

✓
- P.20

Phase 14 — Deep Learning & LLM Foundations
Neural nets, transformers, and how LLMs work

✓
- P.21

Phase 15 — Prompt Engineering & GenAI App Dev
Building a working LLM-powered application

✓
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Building enterprise knowledge-retrieval systems

✓
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Phase 17 — Agentic AI & Frameworks
LangChain, LangGraph, CrewAI, AutoGen — the program's core focus

✓
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Phase 18 — Open Source LLMs & Model Optimization
Local deployment with Ollama and quantization

✓
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Phase 19 — MLOps, LLMOps & Cloud AI
Deploying and operating AI systems in production

✓

SECTION IV — CAPSTONE & CAREER OUTCOMES

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Capstone & Portfolio Projects
10 DevOps projects + 3 enterprise AI capstone tracks

✓
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Tools Mastered & Career Outcomes
Full tooling matrix across 12 categories and 9 target job roles

✓
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Official Enrollment, Admission & Directory
Admission roadmap, cohort schedules, fees & contact

✓

CENTERPIECE MODULE FOCUS

Phase 17: Agentic AI Systems & Frameworks is the program's centerpiece (19 Hours) — the single largest AI phase, backed by 180 hours of real infrastructure to deploy it on.

The Complete DevOps, MLOps & AI Engineering Program Upgrades

The MLOps Engineering Program upgrades working infrastructure professionals — and complete beginners — into engineers who can build, deploy, and operate both traditional and AI-powered systems.

PROGRAM OVERVIEW

This program fuses two GRRAS tracks into one deliberate sequence: 180 hours of true DevOps & Cloud Infrastructure foundations, followed immediately by 100 hours of Machine Learning & Generative AI — with an emphasis on Generative and Agentic AI application development, the highest-value, fastest-growing skill area in tech. Every AI module reuses tools and infrastructure built earlier in the program, so nothing is learned in isolation.

Audience & Prerequisites

- ✓ No prior Linux, cloud, scripting, Python, or ML background is required — the program starts from true zero in both halves.
- ✓ Working professionals already in IT/helpdesk/sysadmin roles can join directly at a faster pace given existing exposure.
- ✓ Developers who want infrastructure and AI deployment ownership over their own applications.
- ✓ Anyone targeting DevOps, Cloud, SRE, Platform, MLOps, or Agentic AI Engineer roles in one program instead of two separate courses.

280-Hour Time Weighting Distribution

Linux, Networking & Systems Foundations	16%
RHCSA-level Linux administration, server management, and networking	
Automation & Infrastructure as Code	12%
Ansible automation and Terraform-driven provisioning	
Cloud Platforms & Virtualization	11%
AWS core services plus virtualization fundamentals	
Containers & Orchestration	12%
Docker and production-grade Kubernetes	
CI/CD, Observability & Source Control	11%
Jenkins, GitHub Actions, Prometheus, Grafana, Git/GitHub	
Data Science & Classic ML Foundations	8%
Python for AI, EDA, and a trained ML model	
Generative & Agentic AI (Core Focus)	21%
LLM fundamentals, RAG, LangChain/LangGraph, CrewAI, AutoGen	
MLOps / LLMOps & Cloud AI	4%
Deploying and monitoring AI systems in production	
Capstone Projects (DevOps + AI Portfolio)	4%
10 DevOps projects plus 3 enterprise AI capstone tracks	

Why This Program, Why Now?

AI is no longer a separate department — it's wired directly into the infrastructure, pipelines, and platforms that DevOps and Cloud engineers already own. Engineers who only know containers and clusters will be managed by engineers who also know how to build, deploy, and operate AI systems on top of them.

EXECUTIVE PROGRAM MANDATE

“This is not two courses stapled together. It's one structured, sequential, 280-hour pathway that takes a complete beginner — or a working sysadmin — from zero to production-grade infrastructure, and then from zero to production-grade Agentic AI, using the exact same skills, tools, and discipline throughout.”

6 Shifts Driving Demand for Combined DevOps + AI Skills

2026 MARKET DYNAMICS

Cloud-Native Is Now the Baseline

01

Docker, Kubernetes, and Infrastructure as Code are no longer differentiators — they're the minimum bar for almost every mid-to-senior engineering role.

Cloud Giants Are Hiring AI Engineers

02

AWS, Azure, and GCP have each launched managed GenAI platforms (Bedrock, Azure AI Studio, Vertex AI) and are actively hiring engineers who can operate them — on top of the core cloud skills this program teaches first.

Explosive Demand for LLMOps & MLOps Engineers

03

“MLOps Engineer” and “AI Infrastructure Engineer” are among the fastest-growing job titles, and nearly every posting expects a solid DevOps/Cloud foundation already in place.

Production Is the New Frontier

04

Organizations are moving from experimenting with AI to running it in production — requiring engineers who understand infrastructure *and* AI, not data scientists alone.

Agentic AI Is the Next Platform Shift

05

Autonomous systems that reason, plan, and take action are recognized as the next major platform shift after cloud and containers — and they still need to be deployed, secured, and monitored like any other system.

Observability Is Becoming Predictive

06

The same Prometheus/Grafana skills taught in this program's DevOps half are the exact substrate AIOps and self-healing infrastructure are built on.

CAREER INSIGHT: DO NOT GET LEFT BEHIND

Classical DevOps automation (Bash, Jenkins, Ansible) is rapidly being augmented with self-healing, agentic AI loops. Engineers who master both infrastructure *and* agentic frameworks command the highest compensation bands in tech.

Who Should Enroll & Key USPs

Engineered for absolute beginners, career switchers, and working infrastructure professionals who want DevOps *and* AI in one program.

Target Audience & Profile Alignment

Complete beginners & career switchers targeting Cloud/DevOps/SRE/AI roles with zero prior technical background.

System administrators moving from traditional IT/helpdesk work into modern, AI-augmented cloud operations.

Existing DevOps/Cloud engineers who want to add AI, ML, and agentic systems to their toolkit without starting a separate course.

Developers who want full ownership of deployment, infrastructure, and the AI layer of their own applications.

ENTRY GUARANTEE
No prior Linux, cloud, Python, or ML experience required. Every subject — from your first Linux command to your first deployed AI agent — is taught from the ground up, in order.

10 Differentiators Why This Program Leads the Market

GRRAS EXCLUSIVE

One Program, Two Disciplines
180 hours of real DevOps + 100 hours of real AI — sequenced so every AI lab deploys onto infrastructure you've already built, not a toy sandbox.

Zero-to-Production Curriculum
Starts at “what is an IP address” and ends with a deployed, containerized, CI/CD-driven Agentic AI application running on your own Kubernetes cluster.

Agentic AI Core Focus
19 of the 100 AI hours are dedicated purely to autonomous agents and multi-agent frameworks — the single most in-demand GenAI skill in tech.

100% Hands-On Learning
31+ modules, 31+ hands-on labs, and 13 portfolio-grade projects. Every concept is immediately applied, never just theory.

Full-Stack Tooling
Linux, Ansible, Terraform, Docker, Kubernetes, Jenkins, Prometheus/Grafana — plus LangChain, LangGraph, CrewAI, AutoGen, RAG pipelines, and multi-provider LLM APIs.

AI Woven Into Every DevOps Tool
Every DevOps module includes an explicit “AI/MLOps Connection” callout showing exactly how that tool is reused later to run AI systems.

DevOps Discipline Applied to AI
Teaches MLOps & LLMOps as an extension of classic DevOps — containerizing, deploying, monitoring, and CI/CD-ing AI systems, not just building them.

Flexible Pacing Options
10 hrs/week (28 wks), 8 hrs/week (~35 wks), or 5 hrs/week weekend format to fit around working professionals' schedules.

Red Hat-Aligned Depth
RHCSA/RHCE-grade Linux and Ansible training — not surface-level command memorization.

GRRAS Brand Trust
Powered by Partnerships with Red Hat & The Linux Foundation
From India's trusted DevOps & Cloud training leader — backed by GRRAS's track record of training 25,000+ engineers.

Program at a Glance & 19-Phase Flow

A step-by-step 280-hour progression building from foundational Linux & networking up to production-grade Agentic AI.

PHASE	TITLE & FOCUS	TRACK	LEARNING OUTCOME
Phase 1	DevOps Foundations & Networking	DevOps	Understand the DevOps lifecycle, toolchain landscape, and core networking underlying every later tool.
Phase 2	Linux Administration (RHCSA-Level)	DevOps	Full command of Linux system administration: users, permissions, storage, services, security, boot process.
Phase 3	Server Management	DevOps	Understand server types, protocols, and the client-server model behind every service you'll run.
Phase 4	Automation with Ansible (RHCE-Level)	DevOps	Automate infrastructure at scale using playbooks, roles, and dynamic inventories.
Phase 5	Source Code Management (Git & GitHub)	DevOps	Master Git workflows, branching, and collaborative review used in every CI/CD pipeline.
Phase 6	Virtualization & Cloud Computing	DevOps	Understand virtualization fundamentals and cloud computing models across AWS, Azure, GCP.
Phase 7	AWS Cloud (Amazon Web Services)	DevOps	Deploy, secure, and manage real workloads across AWS's core compute, storage, network & IAM services.
Phase 8	Terraform (Infrastructure as Code)	DevOps	Provision and manage cloud infrastructure declaratively and repeatably across multiple providers.
Phase 9	Containerization with Docker	DevOps	Package, network, and orchestrate applications as containers, from Dockerfile to Compose.
Phase 10	Kubernetes	DevOps	Deploy and operate production-grade container orchestration, from pods to RBAC to ingress.
Phase 11	CI/CD with Jenkins & GitHub Actions	DevOps	Build pipelines that automatically build, test, and deploy applications.
Phase 12	Monitoring with Prometheus & Grafana	DevOps	Instrument infrastructure with real-time metrics, dashboards, and alerting.
Phase 13	Data Science & ML Foundations	AI/GenAI	Python for AI, exploratory data analysis, and a trained ML model — built from zero.
Phase 14	Deep Learning & LLM Foundations	AI/GenAI	Master how neural networks, transformers & LLMs actually work under the hood.
Phase 15	Prompt Engineering & GenAI App Dev	AI/GenAI	Build a working LLM-powered application from scratch using modern API design.
Phase 16	Embeddings, Vector Databases & RAG	AI/GenAI	Construct a full RAG pipeline — building a private enterprise 'Company GPT'.
Phase 17	Agentic AI & Frameworks (Core Focus)	AI/GenAI	Autonomous single- & multi-agent systems using LangChain, LangGraph, CrewAI & AutoGen.
Phase 18	Open Source LLMs & Model Optimization	AI/GenAI	Run, quantize, and benchmark Llama, Mistral & DeepSeek models locally with Ollama.
Phase 19	MLOps, LLMOps & Cloud AI	AI/GenAI	CI/CD pipelines, containerized deployments, and real-time observability for AI systems.
Capstone	DevOps Portfolio + AI Enterprise Tracks	Capstone	10 DevOps projects + 3 deployed, portfolio-ready Agentic AI systems on cloud infrastructure.

KEY HIGHLIGHT

Phase 2 (Linux) is the largest DevOps phase, and Phase 17 (Agentic AI) is the largest AI phase — together they anchor the program's identity: deep systems fundamentals, paired with the most in-demand AI skill in tech.

Culture, toolchain, and the networking fundamentals every other phase depends on.

Mod 1

The DevOps / AI Landscape

TOPICS COVERED

- DevOps vs AI vs MLOps — clear definitions and where each discipline begins and ends
- The infrastructure lifecycle: provision → configure → deploy → monitor → automate
- Toolchain overview: CI/CD, version control, cloud platforms, containerization & monitoring
- Real industry use cases: automated deployments, self-healing infrastructure, AI-augmented operations
- Where an 'AI-Ready DevOps Engineer' sits relative to classic sysadmin and pure data-science roles

TOOLS & ECOSYSTEM

Conceptual module

HANDS-ON LAB PROJECT

Workshop: map a real company's software delivery process onto the DevOps lifecycle and identify where automation — and later, AI — plugs into each stage.

AI / MLOPS CONNECTION

Every tool you learn from here forward has an AI-augmented counterpart — from AI code-review bots in Git to AI-driven incident triage in monitoring. You'll spot exactly where each one fits as you go.

Mod 2

Networking for DevOps & AI Workloads

TOPICS COVERED

- Networking fundamentals and why they matter for every service you'll run
- IP addressing: IPv4/IPv6, static vs dynamic, address ranges, public vs private
- MAC addresses and their role in network communication
- TCP vs UDP: protocols, characteristics, and use cases
- The OSI & TCP/IP models — all seven layers and how requests actually travel

TOOLS & ECOSYSTEM

CLI networking tools

HANDS-ON LAB PROJECT

Diagram a request's full path through the OSI model and troubleshoot a simulated connectivity failure.

AI / MLOPS CONNECTION

This exact request path is what you'll trace later when debugging why an app can't reach an LLM API endpoint, or why a vector database query is timing out.

The single largest phase in the program — deep, production-grade Linux fundamentals.

Mod 4

Linux Administration (RHCSA-Level)

TOPICS COVERED

- Authentication, authorization & auditing; user creation and management
- Permissions: basic, special, ACLs, and attributes
- Process management & monitoring via GUI and CLI (ps, top, htop, vmstat, iostat)
- RPM packages and repository concepts; installing software and patching
- Service management & configuration using systemctl
- Networking: IP configuration via GUI & CLI, troubleshooting network issues
- Storage management: partitioning with fdisk/parted, swap, LVM, Thin Provisioning, VDO, Stratis, fstab
- Shell scripting for administration automation
- Task scheduling with cron and anacron
- Log management & auditing with auditctl and journalctl
- Backup strategies and remote data transfer
- Encryption & SSH: secure access, passwordless authentication, SSH hardening
- Boot process, troubleshooting boot issues, and root password recovery
- Run levels and targets; performance tuning using static tuning profiles

TOOLS & ECOSYSTEM

Linux (RHEL-family) systemctl fdisk / parted cron journalctl SSH

HANDS-ON LAB PROJECT

Provision and fully harden a Linux server end-to-end — users, permissions, storage, scheduled jobs, SSH hardening, and boot recovery — on a live terminal environment.

AI / MLOPS CONNECTION

Every production LLM server — whether serving Llama locally via Ollama or running a RAG pipeline — is, underneath, a Linux box you must secure, monitor, and tune exactly like this module teaches. GPU driver installs, Python environments, and AI runtime dependencies all live at this layer too.

Understand server roles and the client-server model powering every service.

Mod 3

Server Management

TOPICS COVERED

- Introduction to servers: physical, virtual, and cloud-based
- Client-server architecture and interactions
- Services, daemons, ports, and sockets
- Protocols overview: HTTP, HTTPS, FTP, DNS
- Server types & roles — Web (Apache, Nginx), DNS (BIND), Database (MySQL, MariaDB), File (NFS)

TOOLS & ECOSYSTEM

Apache Nginx BIND MySQL / MariaDB NFS

HANDS-ON LAB PROJECT

Stand up and configure a web server and a file server, and verify client-server communication end-to-end.

AI / MLOPS CONNECTION

The client-server model taught here is exactly how your app's backend will later call an LLM API server — request in, response out, over the same ports and protocols.

Mod 5

Automation Using Ansible (RHCE-Level)

TOPICS COVERED

- Understanding automation and installing Ansible
- Ad-hoc commands for administrative tasks
- Playbooks: writing and running complex automation workflows
- Variables, secrets, and facts
- Task control: loops and conditionals
- Handlers and error handling
- Jinja2 templating for configuration management
- Editing configuration files with file management modules
- Dynamic inventories and parallelism for large projects
- Include/import tasks within playbooks
- Roles: creation and deploying via Ansible Galaxy
- Troubleshooting playbooks and hosts
- Automating provisioning of AI/ML infrastructure — Python environments, model dependencies, GPU drivers — using the same playbook patterns

TOOLS & ECOSYSTEM

Ansible

Jinja2

Ansible Galaxy

HANDS-ON LAB PROJECT

Build a multi-role Ansible project that provisions, configures, and validates a fleet of servers from a single playbook run.

AI / MLOPS CONNECTION

The playbooks you write here are the direct ancestor of the 'agentic DevOps' playbooks used later in the program, where an AI agent reads alerts and decides which Ansible task to trigger for remediation.

Mod 6

Source Code Management: Git & GitHub

TOPICS COVERED

- Version control systems: centralized vs distributed
- Why Git: speed, collaboration, and change tracking
- Git workflows: local and remote repositories
- Git fundamentals: clone, add, commit, push, pull, pull requests, approvers, webhooks
- Branching strategies: feature branches, merging, conflict resolution
- Version tagging with semantic versioning

TOOLS & ECOSYSTEM

Git

GitHub

HANDS-ON LAB PROJECT

Simulate a team workflow — feature branches, a pull request with review comments, a merge conflict, and a tagged release.

AI / MLOPS CONNECTION

Prompt engineering without version control is how teams lose track of what actually worked. The Git discipline learned here becomes 'PromptOps' discipline later — versioning prompts, model configs, and agent behavior alongside code.

Virtualization & Cloud Computing

Virtualization fundamentals and the cloud computing models built on top of them.

Cloud

Mod 7

Virtualization & Cloud Computing

TOPICS COVERED

- Virtualization technology, types, and industry need
- Datacenter concepts
- Hypervisors and virtual machine creation/management
- VM snapshots
- Cloud computing models and providers: AWS, Azure, GCP

TOOLS & ECOSYSTEM

Hypervisor platforms

AWS / Azure / GCP overview

HANDS-ON LAB PROJECT

Provision a virtual machine, snapshot it, restore from snapshot, and map the exercise onto equivalent cloud VM concepts.

AI / MLOPS CONNECTION

GPU-backed virtual machines are the base unit of AI compute. The same hypervisor and snapshot concepts you learn here apply directly — just with a different, more expensive, resource profile.

AWS Cloud (Amazon Web Services)

Deep, hands-on AWS — the most in-demand public cloud skill set.

Cloud

Mod 8

AWS Cloud (Amazon Web Services)

TOPICS COVERED

- On-premises vs AWS Cloud; AWS architecture overview
- Compute: EC2 introduction and management
- Networking services and VPC configuration
- DNS with Route 53
- Storage: S3 & Glacier
- Database services
- Identity & Access Management (IAM)
- Monitoring & auditing with Amazon CloudWatch
- Analytical, migration, application, and deployment/management services
- Enterprise application patterns; disaster recovery; business essentials
- Troubleshooting AWS environments
- Overview of AWS's managed AI services (Bedrock, SageMaker) as extensions of the EC2/S3/IAM skills built in this module

TOOLS & ECOSYSTEM

AWS EC2

VPC

Route 53

S3 / Glacier

IAM

CloudWatch

HANDS-ON LAB PROJECT

Design and deploy a secured VPC with an EC2 instance, an S3-backed storage layer, and IAM roles following least-privilege principles.

AI / MLOPS CONNECTION

Every AWS AI service you'll touch later — Bedrock, SageMaker — sits directly on top of the EC2, VPC, IAM, and S3 foundations you build here. Skipping this module means learning AI on cloud services you don't actually understand.

Terraform (Infrastructure as Code)

Declarative, repeatable infrastructure provisioning across cloud providers.

IaC

Mod 9

Terraform (Infrastructure as Code)

TOPICS COVERED

- Introduction to Infrastructure as Code
- Getting started with Terraform; Terraform basics and state
- Working with Terraform across AWS, Azure, and GCP
- Remote state management
- Terraform provisioners
- Import, tainting resources, and debugging
- Terraform modules
- Terraform functions and conditional expressions
- Provisioning GPU instances and vector-database infrastructure declaratively

TOOLS & ECOSYSTEM

Terraform

AWS / Azure / GCP providers

HANDS-ON LAB PROJECT

Write a modular Terraform configuration that provisions a repeatable multi-resource cloud environment with remote state.

AI / MLOPS CONNECTION

The Terraform modules you write here are reused later to stand up RAG and agentic AI infrastructure — the syntax doesn't change, only the resources being provisioned (GPU instances, vector DB clusters, model endpoints).

Introduction to Containerization: Docker

Package and run applications as portable, reproducible containers.

Containers

Mod 10

Introduction to Containerization: Docker

TOPICS COVERED

- Containerization: introduction, industry use cases, and benefits
- Monolithic vs microservices architecture
- Bare metal vs VM vs container deployment
- Docker architecture and installation
- Core Docker commands; building and running containers from images
- Docker Hub and image registries
- Dockerfile and image creation
- Docker networking
- Docker volumes and persistent storage
- Docker Compose for multi-tier applications
- Conceptual introduction to OpenShift Container Platform
- Containerizing Python/AI dependencies, including GPU-enabled base images

TOOLS & ECOSYSTEM

Docker

Docker Hub

Docker Compose

OpenShift (overview)

HANDS-ON LAB PROJECT

Containerize a multi-tier application (app + database) using a custom Dockerfile and Docker Compose, with a persistent volume.

AI / MLOPS CONNECTION

Every LLM application built in the AI phase of this program gets shipped as a Docker container using exactly this module's patterns — Dockerfile, image, registry, Compose.

Kubernetes

Containers

Production-grade container orchestration — the backbone of modern deployment.

Mod 11

Kubernetes

TOPICS COVERED

- Kubernetes architecture and core components
- Setting up a production-ready cluster with network plugins
- kubectl fundamentals
- Pods: architecture, networking, and deployment via commands & definition files
- Deployment strategies: Pod, ReplicationController, ReplicaSet, Deployment, DaemonSet
- Labels and selectors
- Scaling applications and resource limits
- Namespaces
- RBAC: roles and cluster roles
- Exposing applications via Services
- Certificates for resource authentication
- Network policies
- Ingress configuration
- Monitoring cluster events and alerts
- Troubleshooting resources and clusters
- Concepts for serving models at scale and autoscaling GPU-backed workloads (conceptual introduction)

TOOLS & ECOSYSTEM

- Kubernetes
- kubectl
- Ingress controllers
- RBAC

HANDS-ON LAB PROJECT

Deploy a scaled, namespaced application with RBAC-controlled access and an Ingress-exposed endpoint on a live cluster.

AI / MLOPS CONNECTION

The cluster you deploy to in this module's lab becomes the deployment target for the agentic AI systems built later in the program — same kubectl commands, same RBAC model, same ingress pattern.

CI/CD Using Jenkins & GitHub Actions

Delivery

Automated build, test, and deployment pipelines.

Mod 12

CI/CD Using Jenkins & GitHub Actions

TOPICS COVERED

- Continuous Integration: concepts and importance
- CI/CD pipeline stages: building, testing, deploying
- Jenkins installation and configuration
- Jenkins plugins and extending functionality
- Job creation: Freestyle and Pipeline jobs
- Jenkins security setup
- Pipeline scripting: Declarative and Scripted
- Integrations: Slack, email notifications
- GitHub Actions as a CI/CD alternative/complement
- CI/CD patterns for testing and rolling out prompt/model changes safely

TOOLS & ECOSYSTEM

- Jenkins
- GitHub Actions
- Slack integrations

HANDS-ON LAB PROJECT

Build a Jenkins pipeline that automatically builds, tests, and deploys an application to a containerized environment, with Slack notifications on failure.

AI / MLOPS CONNECTION

The exact same build-test-deploy pipeline you construct here becomes the CI/CD backbone for shipping LLM application updates and agent behavior changes safely in production.

Mod 13

Monitoring Tools: Prometheus & Grafana

TOPICS COVERED

- Prometheus: real-time metrics collection from applications and infrastructure
- PromQL: querying and filtering metrics
- Custom alert rules for proactive issue detection
- Grafana: building dynamic dashboards
- Connecting Grafana to Prometheus and other data sources
- Visual alerts and custom reporting
- Foundations for monitoring LLM-specific metrics: latency, token cost, and hallucination rate, layered on top of standard infra metrics

TOOLS & ECOSYSTEM

Prometheus

Grafana

PromQL

HANDS-ON LAB PROJECT

Instrument a running application/cluster with Prometheus metrics and build a Grafana dashboard with at least one custom alert rule.

AI / MLOPS CONNECTION

This is the direct prerequisite for AIOps. Using these same dashboards, later in the program you'll add an AI agent that reads Grafana alerts and proposes a remediation runbook automatically.

Mod 14

The AI / ML / Data Science Landscape

TOPICS COVERED

- AI vs ML vs Deep Learning vs Generative AI — clear definitions and boundaries
- The AI/ML project lifecycle: data → train → evaluate → deploy → monitor
- Training vs Inference — compute, cost, and infrastructure implications
- Real industry use cases: support automation, fraud detection, personalization, code copilots, DevOps copilots
- Where 'AI Engineering' sits relative to the DevOps skills built in Section II

TOOLS & ECOSYSTEM

Conceptual module

HANDS-ON LAB PROJECT

Workshop: map a real company use case onto the AI lifecycle stages and identify the DevOps touch-points at each stage.

Mod 15

Requires: Module 4 (Server Management)

Python for AI Engineers

TOPICS COVERED

- Fast-paced Python refresh: data types, control flow, functions, error handling
- File handling: reading/writing text, CSV, and JSON
- The requests module — calling REST APIs, headers, auth, response handling
- JSON parsing and serialization for AI payloads
- FastAPI basics — building a minimal API endpoint
- Writing automation scripts (e.g. log parsers, batch processors)

TOOLS & ECOSYSTEM

Python 3

requests

json

FastAPI

HANDS-ON LAB PROJECT

Build a small FastAPI service that accepts a JSON payload and returns a processed response; write a script that pulls data from a public API and persists it to disk.

Mod 16

Requires: Mod 15

Data Handling & Exploratory Data Analysis

TOPICS COVERED

- NumPy arrays and vectorized operations
- Pandas DataFrames: loading, filtering, grouping, merging
- Handling missing data, duplicates, and outliers
- Data visualization for EDA (Matplotlib / Seaborn)
- Feature types: numerical, categorical, text, time-series

TOOLS & ECOSYSTEM

NumPy

Pandas

Matplotlib / Seaborn

HANDS-ON LAB PROJECT

EDA notebook on a real-world dataset (e.g. infrastructure/server logs) producing a cleaned dataset and summary visuals.

Mod 17

Requires: Mod 16

Machine Learning Foundations

TOPICS COVERED

- Supervised learning: regression vs classification
- Unsupervised learning: clustering and dimensionality reduction (overview)
- Train/test split, cross-validation, overfitting vs underfitting
- Core algorithms overview: linear/logistic regression, decision trees, random forests, k-means
- Evaluation metrics: accuracy, precision/recall, F1, RMSE, confusion matrix
- Feature engineering basics

TOOLS & ECOSYSTEM

scikit-learn

Pandas

HANDS-ON LAB PROJECT

Train and evaluate a classification model end-to-end — e.g. predicting incident/failure likelihood from infrastructure logs.

Mod 18

Requires: Mod 17

Deep Learning Concepts

TOPICS COVERED

- Neurons, layers, weights, activation functions
- Forward pass, loss functions, backpropagation, gradient descent — intuition, not derivations
- CNNs (vision) and RNNs/LSTMs (sequence) at a high level
- Why Transformers replaced RNNs for language tasks
- GPUs/accelerators and why deep learning needs them

TOOLS & ECOSYSTEM

PyTorch (or TensorFlow/Keras)

HANDS-ON LAB PROJECT

Build and train a simple feed-forward neural network on a small dataset (e.g. digit classification).

Mod 19

Requires: Mod 18

LLM Fundamentals

TOPICS COVERED

- What are LLMs: pretraining, fine-tuning, RLHF (overview)
- Tokens & context windows — tokenization, context limits, cost implications
- How LLMs generate: next-token prediction, temperature, sampling
- Hallucination — why it happens, detection strategies, mitigation techniques
- Structured output generation: JSON mode, function/tool-calling basics
- AI agents basics — what makes an 'agent' different from a chatbot
- System prompting: role, instructions, guardrails
- Model landscape: OpenAI, Anthropic (Claude), Google (Gemini), open-source LLMs — how to choose

TOOLS & ECOSYSTEM

OpenAI SDK

Anthropic SDK

HANDS-ON LAB PROJECT

Call an LLM API to summarize infrastructure log data and return a structured JSON incident report.

Mod 20

Requires: Mod 19

Prompt Engineering Mastery

TOPICS COVERED

- Prompt frameworks (task – context – format, chain-of-thought style structuring)
- Role-based prompting (system vs user roles)
- Few-shot prompting: example selection and formatting
- Structured prompting for consistent, parseable output
- AI workflow design — chaining prompts for multi-step tasks
- Real business prompting patterns (support, HR, DevOps, infrastructure use cases)

TOOLS & ECOSYSTEM

Anthropic / OpenAI SDKs

Prompt templates

HANDS-ON LAB PROJECT

Design and iteratively test a prompt framework against a set of edge cases.

Mod 21

Requires: Mod 20

Building GenAI Apps with LLM APIs

TOPICS COVERED

- Application architecture: client → backend → LLM API → response
- Multi-provider integration patterns (OpenAI, Anthropic, Google, open-source via API)
- Tool / function-calling fundamentals
- Error handling, rate limits, retries, and cost control for LLM apps
- Logging and basic observability for LLM calls

TOOLS & ECOSYSTEM

FastAPI

OpenAI / Anthropic / Google SDKs

HANDS-ON LAB PROJECT

Build a mini HR Assistant, DevOps Assistant, Resume Analyzer, or Infrastructure Assistant as a FastAPI-backed LLM app.

Embeddings, Vector Databases & RAG

GenAI

Construct a full RAG pipeline — building a private enterprise 'Company GPT' system.

Mod 22

Requires: Mod 19

Embeddings & Vector Databases

TOPICS COVERED

- Embeddings — what they represent, embedding models
- Semantic search vs keyword search
- Similarity search: cosine similarity, ANN indexes (overview)
- Chunking strategies: fixed-size, semantic, overlap
- Metadata filtering for targeted retrieval
- Vector DB landscape: ChromaDB, Pinecone, FAISS — when to use which

TOOLS & ECOSYSTEM

ChromaDB

FAISS

Pinecone

Embeddings API

HANDS-ON LAB PROJECT

Build a Knowledge Search System — embed a document set and return top-k relevant chunks for a query.

Mod 23

Requires: Mod 22

RAG Architecture & Retrieval Pipeline

TOPICS COVERED

- RAG architecture end-to-end: ingestion → chunking → embedding → retrieval → context injection → generation
- Document processing pipelines (PDF, docs, structured data)
- Retrieval pipeline tuning: top-k, re-ranking, hybrid search
- Context injection strategies and prompt construction for RAG
- Hallucination reduction techniques specific to RAG
- Enterprise knowledge system design patterns

TOOLS & ECOSYSTEM

ChromaDB / FAISS

LLM APIs

Document loaders

HANDS-ON LAB PROJECT

Build a Private Company GPT, DevOps Knowledge Assistant, or Internal SOP Chatbot.

Mod 24

Requires: Mod 21

AI Agents — Foundations

TOPICS COVERED

- What makes something an 'agent' (goal-directed, tool-using, iterative)
- The agent loop and ReAct-style reasoning
- Tool / function calling as the agent's 'hands'
- Short-term vs long-term memory in agents
- Planning strategies: single-step, multi-step, plan-and-execute
- Failure modes — loops, tool misuse, cost runaway — and guardrails

TOOLS & ECOSYSTEM

LLM APIs with function/tool calling

HANDS-ON LAB PROJECT

Build a single-tool agent — e.g. an agent that queries a log file and answers questions about it.

Mod 25

Requires: Mod 24

LangChain & LangGraph for Agentic Workflows

TOPICS COVERED

- LangChain fundamentals: prompt templates, chains, output parsers
- Memory in LangChain (conversation buffer, summary memory)
- Retrieval pipelines integrated into chains (RAG + LangChain)
- Tool calling within LangChain
- Introduction to LangGraph: state graphs, conditional edges, cycles
- AI automation workflow design patterns

TOOLS & ECOSYSTEM

LangChain

LangGraph

HANDS-ON LAB PROJECT

Multi-Step AI Assistant — a chained workflow with memory, retrieval, and at least one tool call.

Multi-Agent Systems & CrewAI

CrewAI & AutoGen orchestration, plus DevOps-native agentic applications.

Agentic AI

Mod 26

Requires: Mod 25

Multi-Agent Frameworks — CrewAI & AutoGen

TOPICS COVERED

- Why multi-agent: role specialization, parallelism, critique/verification patterns
- CrewAI: agents, tasks, crews, sequential vs hierarchical processes
- AutoGen: conversable agents, group chat, human-in-the-loop patterns
- Orchestration patterns: manager/worker, debate/critique, pipeline
- Cost, latency, and reliability trade-offs of multi-agent systems

TOOLS & ECOSYSTEM

CrewAI AutoGen

HANDS-ON LAB PROJECT

Build a 2–3 agent crew (e.g. researcher + writer + reviewer) that completes a shared task.

Mod 27

Requires: Mod 26

Agentic DevOps Applications

TOPICS COVERED

- DevOps-specific agent use cases: incident triage, log analysis, runbook automation, infra provisioning assistants
- Wiring agents to real tools: shell/CLI wrappers, cloud SDKs, ticketing/monitoring APIs as callable tools
- Human-in-the-loop approval steps for high-risk actions
- Evaluating agent output: correctness, safety, cost, and observability

TOOLS & ECOSYSTEM

LangChain / CrewAI Cloud/infra APIs (simulated or sandboxed)

HANDS-ON LAB PROJECT

Build a DevOps/Infrastructure Assistant agent — e.g. an agent that reads alerts, checks logs, and proposes a remediation runbook, using the Ansible and monitoring skills from Section II.

Open Source LLMs & Model Optimization

Run, quantize, and benchmark open models locally with Ollama.

Open LLMs

Mod 28

Requires: Mod 19

Open Source LLMs & Local Deployment

TOPICS COVERED

- Local AI setup: Ollama / local inference servers
- Running open-source models: Llama, Mistral, DeepSeek, Gemma
- LLM comparison: quality, cost, latency, licensing considerations
- Choosing open-source vs hosted APIs: data residency, cost, latency, control
- Model optimization basics: quantization, distillation (overview), LoRA (overview)

TOOLS & ECOSYSTEM

Ollama Hugging Face Open-source models

HANDS-ON LAB PROJECT

Deploy an open-source model locally and benchmark it against a hosted API on the same task, on the Linux server hardened in Phase 2.

MLOps, LLMOps & Cloud AI

CI/CD pipelines, containerized deployments, and real-time observability for AI systems — the direct continuation of Section II's DevOps discipline.

Mod 29

Requires: Mod 17

MLOps Fundamentals

TOPICS COVERED

- ML lifecycle: data → train → validate → deploy → monitor → retrain
- Experiment tracking
- ML pipelines — orchestrating multi-step training/deployment workflows
- Model versioning and registries
- Data versioning

TOOLS & ECOSYSTEM

MLflow DVC Kubeflow (overview) BentoML (overview)

HANDS-ON LAB PROJECT

Track an experiment and version a model using MLflow.

Mod 30

Requires: Mod 29

MLOps & LLMOps in Production

TOPICS COVERED

- AI model deployment patterns
- Kubernetes for AI: serving at scale (overview) — building on Phase 10
- Model monitoring: drift, latency, cost, and quality metrics for LLM apps
- Docker for AI workloads (dependencies, GPU considerations) — building on Phase 9
- API serving with FastAPI + model/LLM backend
- CI/CD for ML models and prompts: testing, versioning, rollout — building on Phase 11

TOOLS & ECOSYSTEM

Docker Kubernetes FastAPI CI/CD (GitHub Actions)

HANDS-ON LAB PROJECT

Containerize and deploy one of the earlier agentic/RAG apps with a basic CI/CD pipeline, onto the Kubernetes cluster from Phase 10.

MLOps, LLMOps & Cloud AI

AI on Cloud — deploying managed LLM/ML services across AWS, Azure, and GCP.

Mod 31

Requires: Mod 30

AI on Cloud

TOPICS COVERED

- AWS: Bedrock basics, SageMaker overview — building on Phase 7
- Azure: Azure OpenAI, AI Studio, Cognitive Services
- Google Cloud: Vertex AI overview
- Choosing managed vs self-hosted: cost, control, compliance trade-offs
- Security & IAM basics for AI services on cloud

TOOLS & ECOSYSTEM

AWS Bedrock / SageMaker Azure AI Studio Vertex AI

HANDS-ON LAB PROJECT

Deploy or call a managed LLM/ML service on at least one cloud provider, using the IAM and VPC skills from Phase 7.

Capstone & Portfolio Projects

A 13-project portfolio spanning both halves of the program — 10 DevOps projects that build your infrastructure, and 3 enterprise AI tracks that deploy directly onto it.

DevOps Portfolio — 10 Projects (Built Throughout Section II)

P1

Server Hardening & Security Optimization
SSH hardening, firewall configuration, and access-control best practices on production Linux systems.

P2

Automated Backup & Disaster Recovery
Automated backup procedures and a disaster recovery plan ensuring business continuity.

P3

Scalable Web Server Infrastructure (Ansible)
Load-balanced, SSL-secured, hardened web hosting deployed entirely via Ansible playbooks.

P4

Automated Storage Management (Ansible)
Automated disk partitioning and storage provisioning across a fleet of servers.

P5

Hosting a Website with Load Balancer (AWS)
High-availability static site hosting behind an AWS Elastic Load Balancer.

P6

Multi-Tier Architecture on AWS
Load Balancer + EC2 application layer + RDS database layer for a scalable, secure app.

P7

Infrastructure Automation with Terraform
Repeatable, secure, multi-resource cloud provisioning entirely as code.

P8

System Monitoring with Prometheus & Grafana
Real-time observability and alerting dashboards for production infrastructure.

P9

Multi-Tier App on Kubernetes + Monitoring
Web + API + DB deployed on Kubernetes with a live Prometheus/Grafana dashboard — **the exact cluster reused by the AI capstone tracks below.**

P10

End-to-End CI/CD for a Java App with Docker
Jenkins pipeline: build → test → Dockerize → deploy across Dev/UAT/Prod — **the exact pipeline pattern reused for AI app deployment.**

Final Program Requirement (6 Hours) — Choose One AI Capstone Track

TRACK A

Autonomous DevOps Incident Response Agent
RAG + agent + tool calling + deployment. Reads alerts, checks logs via RAG, and executes automated runbook remediation scripts on the Kubernetes cluster from Phase 10.

RAG Pipeline

Autonomous Agent

Tool Calling

Kubernetes

FastAPI

TRACK B

Enterprise 'Private GPT' Knowledge Assistant
RAG + agentic follow-up actions + cloud deployment. Secure multi-tenant RAG system with role-based access control and vector search, deployed via the CI/CD pipeline from Phase 11.

FAISS/ChromaDB

LangChain

RBAC Security

Docker Deployment

TRACK C

Multi-Agent Infrastructure Automation System
CrewAI/AutoGen + real/simulated infra tools + monitoring. Specialized AI agents collaborate to audit cloud security and generate Terraform code, using the IaC skills from Phase 8.

CrewAI

AutoGen

Terraform

MLflow Observability

Required Deliverables

- ✓ A working, containerized, deployed application
- ✓ An architecture diagram
- ✓ A short demo plus a write-up of design decisions, evaluation, and known limitations

Evaluation & Assessment

Live project demo, evaluated against a rubric covering functionality, architecture quality, safety/guardrails, and presentation — assessed by instructors from both the DevOps and AI faculty.

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Tools Mastered & Career Outcomes

A full-stack, 12-category tooling matrix spanning infrastructure and AI — plus 9 target job roles this program prepares you for.

Linux & Systems

RHEL-family LinuxBash ScriptingsystemctlSSH

Cloud Platforms

AWS EC2/VPC/S3/IAMRoute 53CloudWatchAzure & GCP (overview)

Data Science / ML

NumPyPandasscikit-learnMatplotlib/Seaborn

Networking

TCP/IPOSI ModelDNSIP Addressing

Containers & Orchestration

DockerDocker ComposeKubernetesOpenShift (overview)

LLM Providers & Agentic Frameworks

OpenAIAnthropicGoogleLangChainLangGraphCrewAIAutoGen

Automation & IaC

AnsibleTerraformJinja2

CI/CD & Observability

JenkinsGitHub ActionsPrometheusGrafana

Vector DBs & Open LLMs

ChromaDBFAISSPineconeOllamaHugging Face

Version Control

GitGitHub

Languages & APIs

Python 3FastAPIrequestsJSON

MLOps / LLMOps / Cloud AI

MLflowDVCAWS Bedrock/SageMakerAzure AI StudioVertex AI

Target Career Specializations & Job Roles

MLOps EngineerHIGHEST DEMAND

AI / LLMOps EngineerHIGH GROWTH

Agentic AI EngineerHIGHEST PAID

GenAI Application DeveloperIN-DEMAND

DevOps Engineer with AI SpecializationCAREER UPGRADE

Cloud AI EngineerENTERPRISE

Site Reliability Engineer (SRE)FOUNDATIONAL

Platform EngineerIN-DEMAND

Cloud / Automation EngineerFOUNDATIONAL

WHY UPGRADE YOUR CAREER AT GRRAS

✓ 25,000+ students already trained in Linux, Red Hat, Cloud & DevOps — this program is the natural, logical next step.

✓ Same trusted instructors and hands-on, lab-first teaching style known across top IT hubs.

✓ One program covering both disciplines the market now hires for together — not two disconnected bootcamps.

✓ Every lab and capstone is infrastructure- and AI-flavored: log analysis, incident response, runbook automation, and deployed agentic systems.

✓ A single, sequential 280-hour learning path — eliminating scattered tutorials and outdated, siloed courses.

ENROLLMENT NOW OPEN

Start the Complete DevOps, MLOps & AI Engineering Program Today

Join India's premier combined DevOps + AI/MLOps program, backed by GRRAS Solutions' 25,000+ student trust legacy. 280 hours. 19 phases. One engineer who can run the infrastructure and the AI on top of it.

4-Step Admission & Onboarding Roadmap

Online Registration

STEP 01

Fill out the application form at www.grras.com or contact an advisor.

Counseling & Prerequisites

STEP 02

Free 1-on-1 profile review with a DevOps/AI lead instructor.

Portal & Lab Access

STEP 03

Receive instant access to Linux pre-work and cloud sandbox environments.

Live Cohort Launch

STEP 04

Begin your 280-hour sequential journey with expert live guidance.

Cohort Schedules & Learning Modes

Weekday Intensive Cohort — Mon–Thu evenings | 10 hrs/week | ~28 weeks

Weekend Executive Cohort — Sat & Sun, flexible | 5 hrs/week | ~56 weeks

Live Interactive Hybrid — Recorded sessions + live Q&A |

Fee Plans & Financing Assistance

- ✓ **Standard One-Time Payment** — includes all course materials, cloud lab credits & certification for the full 280-hour program.
- ✓ **Flexible 0% Interest EMI** — easy monthly installments across the program's DevOps and AI phases, zero hidden charges.
- ✓ **Corporate / Team Training** — special group pricing for teams of 3+ engineers from the same organization.
- ✓ Contact enquiry@grras.com for current combined-program fee details and available discounts.

Verifiable Credential

Receive an official **GRRAS Certified MLOps Engineer** Certificate of Completion, covering both the DevOps and AI/GenAI halves of the program, with a unique QR code for instant LinkedIn and employer verification.

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